Name: ______________________

Worksheet

Calculating integers from their prime factor expressions

TASK A: Evaluating products of prime factors

Write the following as integers:

1) $2^3 \times 3^2 \times 5^2$

2) $5^3 \times 7$

3) $2^2 \times 3 \times 11^2$

4) $3^3 \times 7 \times 13^2$

5) $5^2 \times 7^3 \times 19$

TASK B: Using product of prime factor form

1) Using prime numbers make each of the following products satisfy the condition on the right

a)	$3^2 \times 5^3$	an even number
b)	2^2	a multiple of 6
c)	$2^4 \times 7$	a multiple of 9
d)	$3 \times 5^3 \times 11^2$	a multiple of 10
e)	$3^2 \times 5^3 \times 13$	a square number
f)	$3^2 \times 5^3 \times 11^4$	a cube number

2) Here is a number written as a product of its prime factors

Do not work it out

$$2^4 \times 5^2 \times 7^2$$

Are the following statements true or false make sure you justify your answers?

a) The number is even
b) The number ends in a zero
c) The number ends in two zeros
d) The number has 35 as a factor
e) The number has 50 as a factor
f) It is a square number

3) List all the common factors of:

$2^4 \times 3 \times 7^2$ and $2^3 \times 3^3 \times 7$